

# Regularized Feature Selection Landscapes: An Empirical Study of Multimodality



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**Abstract.** The processing of features in data is among the key topics in machine learning. While a broad range of heuristics for feature processing, including feature selection, have been developed and experimented with, less research has been concerned with the underlying fitness landscape. In this paper, we perform a fitness landscape analysis of feature selection, using local optima networks and other methods. We focus on the impact of regularization, an important element of many machine learning methods. Our study using ten datasets and learning of decision trees confirms and adds to previous findings that feature selection landscapes are highly multimodal. It is the first study to focus on the impact of regularization on the landscape induced by feature selection. In the ten datasets studied, we find a high degree of multimodality when there is no regularization. With increasing regularization, the degree of multimodality generally drops off but remains substantial.

**Keywords:** Machine Learning · Feature Selection · Regularization · Landscape Analysis · Multimodality · Local Optima Networks · Visualization



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